

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _S.Amulya (192321044) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Cloud Auto-Scaling and Load Balancing for Seasonal Traffic Spikes

2.

Introduction

E-commerce companies often experience sudden traffic spikes during seasonal sales such as Black Friday, Diwali, or Amazon Great Indian Festival. Cloud computing provides **auto-scaling** and **load balancing** to maintain application performance and availability during these peak periods.

Auto-Scaling

Auto-scaling automatically increases or decreases computing resources based on application demand.

- During heavy traffic, additional virtual machines or containers are launched automatically.
- When traffic decreases, extra resources are removed to reduce costs.
- This ensures efficient utilization of cloud resources.

Load Balancing

A load balancer distributes incoming user requests evenly across multiple servers.

- Prevents any single server from becoming overloaded.
- Improves response time.
- Ensures continuous availability even if one server fails.

Reducing Downtime and Improving User Experience

- Prevents server crashes during traffic spikes.
- Reduces website response time.
- Maintains uninterrupted shopping experiences.
- Ensures high availability of services.

Role of Monitoring Tools

Monitoring tools continuously track:

- CPU utilization
- Memory usage
- Network traffic
- Response time
- Error rates

If resource usage exceeds predefined thresholds, monitoring tools trigger auto-scaling actions automatically.

Real-World Example

During **Amazon's Great Indian Festival**, millions of users access the website simultaneously. AWS Auto Scaling automatically launches additional EC2 instances, while Elastic Load Balancer distributes customer requests across these servers. CloudWatch monitors system performance and triggers scaling when required.

Conclusion

Cloud auto-scaling, load balancing, and monitoring work together to handle unpredictable traffic efficiently, minimize downtime, and provide a smooth shopping experience.

2. Cloud Storage for Secure Backup and Disaster Recovery

Introduction

Organizations generate large amounts of backup data that must be stored securely and reliably. Cloud storage provides scalable, durable, and cost-effective backup solutions.

Data Durability and Redundancy

Cloud providers ensure data protection by:

- Storing multiple copies of data across different storage devices.
- Replicating data across multiple availability zones or regions.
- Automatically replacing failed storage devices without data loss.

This ensures very high durability and availability.

Encryption

Sensitive backup data is protected using:

- Encryption during transmission (TLS/SSL).
- Encryption while stored (AES-256).
- Secure key management services.

Even if data is intercepted, it cannot be read without the encryption key.

Access Control

Cloud providers implement:

- Identity and Access Management (IAM)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Audit logs

These mechanisms ensure that only authorized users can access backup data.

Cost Advantages

Compared to traditional storage systems:

- No need to purchase expensive storage hardware.
- Pay only for the storage used.
- Reduced maintenance and power costs.

Disaster Recovery Example

A company's on-premise server is damaged during a flood. Since backups are stored in the cloud, the organization restores its applications and data from cloud storage within a short time, minimizing business downtime.

Conclusion

Cloud storage provides secure, highly durable, and cost-effective backup solutions while supporting fast

3. CDN for Global Mobile Application Deployment

Introduction

A mobile application deployed globally requires fast content delivery to users located in different countries. Cloud providers use **Content Delivery Networks (CDNs)** to improve performance.

Content Delivery Network (CDN)

A CDN is a distributed network of servers located in multiple geographical regions.

- Frequently accessed content is cached closer to users.
- Users download data from nearby servers instead of the central server.

Reducing Latency

Latency decreases because:

- Shorter network distance.
- Reduced congestion.
- Faster response times.
- Quicker content delivery.

Role of Edge Servers

Edge servers:

- Store cached copies of application content.
- Process user requests locally.
- Reduce the workload on the origin server.
- Improve application responsiveness.

Global Availability

Cloud providers support global availability through:

- Multiple data centers.
- Global load balancing.
- Automatic failover.
- Geographic redundancy.

Users automatically connect to the nearest available server.

Real-Time Streaming Example

Netflix stores video content on CDN edge servers worldwide. When users watch a movie, the content is delivered from the nearest edge server, reducing buffering and improving streaming quality.

Conclusion

CDNs improve application performance, reduce latency, and provide consistent user experiences worldwide through edge servers and globally distributed cloud infrastructure.

4. Cloud Security, Compliance, and Data Sovereignty in Financial Institutions

Introduction

Financial institutions manage highly sensitive customer information and must comply with government regulations while ensuring strong security.

Regulatory Compliance

Cloud providers support compliance with standards such as:

- ISO 27001
- PCI DSS
- GDPR
- SOC 2

These certifications help organizations meet regulatory requirements.

Logging

Logging records:

- User login activities.

- Data access events.
- System modifications.
- Security incidents.

Logs assist in auditing and forensic investigations.

Monitoring

Monitoring tools continuously detect:

- Unauthorized access attempts.
- System failures.
- Security threats.
- Performance issues.

Alerts enable quick incident response.

Encryption

Encryption protects:

- Data in transit using TLS/SSL.
- Data at rest using AES-256.
- Database backups and storage volumes.

Data Sovereignty

Organizations can:

- Store data within approved geographic regions.
- Select cloud regions that comply with national regulations.
- Restrict cross-border data transfers.

This ensures compliance with local data protection laws.

Financial Transaction Example

A bank stores customer transaction records in an Indian cloud region using encryption. IAM and MFA restrict access to authorized employees, while audit logs record every transaction for regulatory inspections.

Conclusion

Cloud providers help financial institutions achieve compliance through auditing, encryption, monitoring, and regional data storage while protecting sensitive financial information.

5. Containerization and Kubernetes for Application Deployment

Introduction

Modern applications are increasingly deployed using **containers**, which provide lightweight, portable, and efficient application environments. Kubernetes automates the deployment and management of these containers.

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Operating System	Share host OS	Each VM has its own OS
Startup Time	Seconds	Minutes
Resource Usage	Low	High
Performance	Faster	Slightly slower
Portability	Very High	Moderate

Performance and Resource Usage

Containers:

- Require less CPU and memory.
- Start quickly.
- Allow many applications to run on the same server efficiently.
- Improve infrastructure utilization.

Kubernetes Orchestration

Kubernetes automates:

- Container deployment.
- Load balancing.
- Auto-scaling.
- Self-healing by restarting failed containers.
- Rolling updates without downtime.

This simplifies application management and improves reliability.

Portability Across Clouds

Containers package the application with all its dependencies, allowing it to run consistently across:

- Private cloud
- Public cloud
- Hybrid cloud
- Multi-cloud environments

This avoids vendor lock-in and simplifies migration.

Microservices Example

An online shopping application can be divided into separate microservices:

- User Authentication Service
- Product Catalog Service
- Shopping Cart Service

- Payment Service
- Order Management Service
- Notification Service

Each service runs in its own container and is managed by Kubernetes, allowing independent deployment and scaling.

Conclusion

Containers provide lightweight and portable application deployment, while Kubernetes automates management, scaling, and recovery. Together, they enable efficient deployment of modern cloud-native microservices applications.